

LESSON 7-7 • HOMEWORK PACKET • LEVEL 1 • SHEET 1 OF 2**Polygons and Perimeter on the Plane**

Maryland Standard 6.G.A.3 • Learning target – I can find the perimeter and area of a polygon drawn on the coordinate plane.

REMEMBER

- Plot the vertices in order. Connect them to draw the polygon.
- Find each horizontal or vertical side from the coordinates.
- Perimeter** = sum of all sides. **Area** of a rectangle = length $\times$ width.

WORKED EXAMPLE

Find the perimeter and area of the rectangle with vertices $(-3, 2)$, $(4, 2)$, $(4, -1)$, $(-3, -1)$.

Step 1 Top: $3 + 4 = 7$ units. Side: $2 + 1 = 3$ units.

Step 2 $P = 7 + 3 + 7 + 3 = 20$ units. $A = 7 \times 3 = 21$ square units.

Answer $P = 20$ units, $A = 21$ square units

WORD BANK

vertex – a corner point

perimeter – distance around; add all sides

area – space inside; length $\times$ width

polygon – a closed shape with straight sides

PRACTICE

show your work • label units

1. Use the rectangle on the grid.

Step 1 Find AB: same y , so use the x -coordinates.

Step 2 Find BC: same x , so use the y -coordinates.

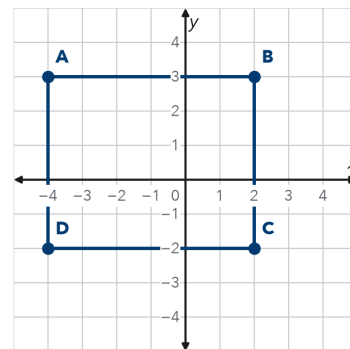
Step 3 Perimeter: **add** 4 sides. Area: **multiply**.

a. Length of AB _____ units

b. Length of BC _____ units

c. Perimeter _____ units

d. Area _____ sq. units



2. Plot $A(-3, -2)$, $B(4, -2)$, and $C(-3, 4)$. Connect them to make a right triangle.

Step 1 Plot the three points and connect them.

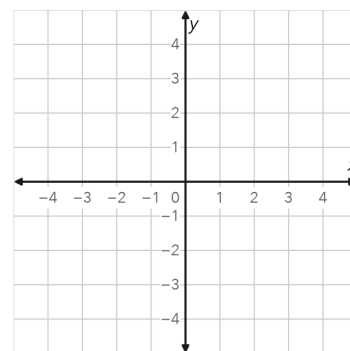
Step 2 Find the base and the height.

Step 3 Area = $\frac{1}{2} \times \text{base} \times \text{height}$.

a. Base AB _____ units

b. Height AC _____ units

c. Area _____ sq. units



- 3. Garden Fence.** A garden has corners at $(-5, 3)$, $(6, 3)$, $(6, -4)$ and $(-5, -4)$. Each unit is 1 meter. How much **fence** goes all the way around it?

Step 1 Find the length of the top side.

Step 2 Find the length of the left side.

Step 3 Add all four sides.

Answer _____ m

- 4. Square.** A square has vertices $(-3, 1)$, $(2, 1)$, $(2, -4)$, and $(-3, -4)$. Find its perimeter.

Answer _____ units

- 5. Reasoning.** Three corners of a rectangle are $(-2, 5)$, $(3, 5)$, and $(3, -1)$. Name the fourth corner.

Frame The fourth corner is _____. It has the same x as _____ and the same y as _____.

Answer _____

CHECK YOURSELF

☐ I can do this on my own

☐ I need a little help

☐ I have a question about #_____

LESSON 7-7 • HOMEWORK PACKET • LEVEL 1 • SHEET 2 OF 2**Designing Polygons with Scale**

Maryland Standard 6.G.A.3 · Learning target – I can use a scale to find a missing dimension of a figure on the coordinate plane.

REMEMBER

- A **scale** tells what 1 grid unit stands for, like 1 unit = 5 ft.
- Real length = grid units $\times$ scale.
- Missing side = area $\div$ known side.

WORKED EXAMPLE

1 unit = 5 ft. A rectangle's sides run from $(-2, 3)$ to $(4, 3)$ and from $(4, 3)$ to $(4, -1)$. Find its real size.

Step 1 Top: $2 + 4 = 6$ units. $6 \times 5 = 30$ ft.

Step 2 Side: $3 + 1 = 4$ units. $4 \times 5 = 20$ ft.

Answer 30 ft by 20 ft

WORD BANK

scale – 1 grid unit = a real length

area – space inside, in square units

dimensions – the length and the width

PRACTICE

show your work · label units

1. **Court Plan.** On this plan, 1 unit = 5 ft. Side AB of a court is drawn. The court must cover 1,200 ft².

Step 1 Count AB in grid units.

Step 2 Multiply by 5 to get feet.

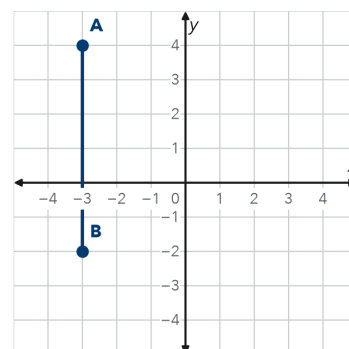
Step 3 Divide 1,200 by that length.

a. AB in grid units _____ units

b. AB in feet _____ ft

c. Other side in feet _____ ft

d. Other side in grid units _____ units



2. On a map, 1 grid unit = 4 meters.

a. 7 units = _____ m

b. 36 m = _____ units

3. **Patio.** A patio on a plan is 7 units by 5 units. **1 unit = 2 ft.** Find the real area.

Step 1 Change each side to feet: multiply by 2.

Step 2 Multiply the two real sides.

Answer _____ ft²

4. **Bedroom.** On a plan, 1 unit = 3 ft. A room has corners at $(-2, 4)$, $(2, 4)$, $(2, -1)$, and $(-2, -1)$. Find its real perimeter.

Step 1 Find each side in grid units.

Step 2 Multiply each side by 3 to get feet.

Step 3 Add all four sides.

Answer _____ ft

5. **Find the Error.** If 1 unit = 3 ft, Lily says $(-4, 1)$ to $(5, 1)$ is 3 ft. Explain her error. Fix it.

Frame Lily should ___ because the signs are ___. The real length is ___ ft.

TEST PRACTICE

choose the best answer

6. A plan uses 1 unit = 6 m. A side is 5 units long. How long is the real side?
- A** 11 m **B** 30 m **C** 1 m **D** 56 m
7. A rectangle has an area of 480 ft^2 . One side is 24 ft. How long is the other side?
- A** 20 ft **B** 456 ft **C** 40 ft **D** 504 ft

CHECK YOURSELF

☐ I can do this on my own ☐ I need a little help ☐ I have a question about #_____
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